CS 2300 HW8

1) According to the question, we can say that:

Base case : 7∈ A

Recursive rule:if s∈A and t∈A, then we can say that st also belongs to A, where st denotes the concatenation of s and t.

From the question, we have B = { 7^i| $i7i|i \in Z^+$}, that represents all positive powers of number 7.

a) **Prove:** We need to prove that every element in A is of the form 7^i for some positive integer i.

Base case: Since 7 ∈ A and $7^1 = 7$, so we can say that the base case holds true.

Inductive step:

Let us assume that S ∈ A, so that means that S = 7^m ∈ B and t ∈ A, that means $7^n \in B$, then by recursive rule we can say that st ∈ A.

Let us say that s and t are some positive integers such s = 777 and t = 77, that means st = 77777, so that means st is still a number consisting only of 7s, that is st = 7^{m+n}, and since m,n ∈ Z^+ , m+n must also belong to Z^+. Thus, st ∈ B

Hence, by structural induction A ⊆ B.

b) Prove: We need to prove that every number of the form 7^i is in A.

Let us use mathematical induction to prove this.

Assume that P(i) : 7^i ∈ A

Base Case (i =1):

7^1 = 7, and 7 belongs to A by rule (1)

Hence, the base case holds true.

Inductive Step:

Assumption: $7^{k} \in A$ for k > 1

Prove: $7^{k+1} \in$ A

$7^{k+1} = 7^{k} * 7$
$7^{k} \in$ A [inductive hypothesis]
$7 \in$ A [Base Case]

Therefore, by concatenation we get $7^{k+1} \in$ A.

2) According to the question, we can say that:

i=0
i=1 : abb
i=2 : aabbbb

Base Case: ε ∈ S (since $a^{0}b^{0}$ = ε)

Recursive step: if w ∈ S, then awbb ∈ S.

So basically starting from ε, the recursive rule prepends an A and appends 2 bs. Each step increases the count of a by 1 and b by 2, which preserves the form $a^{i}b^{2i}$.

3a) Let us assume that D is defined as following:

Base Case: 1 ∈ D, since $2^{0} * 3^{0} * 5^{0}$

The recursive rules for D can be defined as follows:

- If x ∈ D, then 2x ∈ D
- If x ∈ D, then 3x ∈ D
- If x ∈ D, then 5x ∈ D

Therefore, starting from 1, if we apply these rules repeatedly, then we can build any product of 2s,3s and 5s, for example, 1 can be written as $2^{0} * 3^{0} * 5^{0}$, 2 can be written as $2^{1} * 3^{0} * 5^{0}$ and so on and they belong to D as well.

Hence proved.

b) **Base Case:** 1 ∈ D and 1 = $2^{0} * 3^{0} * 5^{0}$, so we can say that the base case holds true.

Inductive Step:

Assumption: if,x ∈ D, then x = $2^k * 3^m * 5^n$ ∈ C

Prove: 2x = $2^{K+1} * 3^m * 5^n$ ∈ C, 3x = $2^K * 3^{m+1} * 5^n$ ∈ C, 5x = $2^K * 3^m * 5^{n+1}$ ∈ C

Over here, we can see that all numbers are clearly in C because increasing the variables k,m and n by 1 and keeping others constant gives an element in the form of $2^k * 3^m * 5^n$.

Hence, by structural induction, we can say that every element of D is in C and that's why D ⊆ C.

4a) A full binary tree is a tree where each internal node has 2 children only.

By inductive definition of full binary trees, we can say that:

Base case: A single vertex is a full binary tree

Recursive Step: If T1 and T2 are full binary trees, then the tree formed by creating a new root and attaching T1 and T2 as its left and right subtrees is also a full binary tree.

Internal vertices: int(T)

Base case: T is a single vertex (a leaf):
→ int(T) = 0

Recursive Case: If tree is built from subtrees T1 and T2, then the new root is an internal node and internal nodes of T are:

int(T) = 1 + int(T1) + int(T2)

b) Total size (T)

T is a single vertex (a leaf):

Base Case:

T is a single vertex (a leaf):
→ size(T) = 1

Recursive Case: If T is built from subtrees T1,T2 then:

size(T): 1 + size(T1) + size(T2)

Proof:

- T is a single leaf node
- int(T) = 0, size(T) = 1

size(T) = 2*0 + 1 = 1
Thus, it holds true for size(T).

Inductive Step:

Let us assume that the property holds true for T1 and T2:

size(T1) = 2 * int(T1) + 1
size(T2) = 2*int(T2) +1

Prove: size(T): 2*int(T)+1

From the above definitions, we can say that:

- int(T) = 1 + int(T1) + int(T2)
- size(T) = 1 + size(T1) + size(T2)

size(T) = 1 + (2 * int(T1) + 1) + (2*int(T2) + 1) = 3 + 2 int(T1) + 2 int(T2)

int(T) = 1 + int(T1) + int(T2) = 2*int(T1) + 1 = 2 * (1 + int(T1) + int(T2) +1) + 1 = 2 * (int(T1) + int(T2)) + 3

Thus, we can say that, size(T) = 2 * int(T) +1

Hence, by structural induction, size(T) = 2 * int(T) + 1 for all binary trees T.

5a) Let us define L' such that:

Base case: (0,0) ∈ L', this is the place where robot starts

Recursive Step: Over here we need to finance rules that define the condition a-b mod 5 = 0, which means that after each and every step x-y mod 5 = 0 must hold true.

Let us define the valid moves that a robot can take:

- (x,y) → (x+5,y) [Move towards right by (5,0)]
- (x,y) → (x-5,y) [Move towards left by (-5,0)]
- (x,y) → (x+1, y+1) [Moves diagonally by (1,1)]

- $(x,y) \rightarrow (x-1,y-1)$ [Moves diagonally by (-1,-1)]

By this way we can preserve the invariant x-y mod 5 = 0 because:

- $(x+5) - y = (x-y) + 5$
- $(x+1) - (y+1) = (x-y)$
- $(x-1) - (y-1) = (x-y)$

Hence, we can say that these steps never change the value of (x-y) mod 5

If L' is the smallest set such that $(0,0) \in L'$, and if $(x,y) \in L'$ then $(a+5,b),(a-5,b),(a+1,b+1),(a-1,b-1)$ also belong to L'.

b) **Prove:** We must prove inductively that $L' \subseteq L$.

Let's define L' by using structural induction.

Base case: Let us take (0,0). Since 0 - 0 =0 and 0 mod 5 = 0 then we can say that $(0,0) \in L'$ and the base case holds true.

Inductive Step:

Assumption: Let us assume that $(x,y) \in L$, which means that x - y mod 5 = 0.

Let us show that each points to a point in L.

$(x+5,y) \rightarrow (x+5)-y=(x-y)+5 = 0+5 = 5$ and 5 mod 5 = 0

$(x-5,y) \rightarrow (x-5)-y=(x-y)-5 = 0-5 = -5$ and -5 mod 5 = 0

$(x+1,y+1) \rightarrow (x+1)-(y+1) = x-y = 0$ and 0 mod 5 = 0

$(x-1,y-1) \rightarrow (x-1)-(y-1)=x-y = 0$ and 0 mod 5 = 0

Hence, we can say that all cases preserve the condition (a - b) mod 5 = 0, so by structural induction $L' \subseteq L$.

c) **Prove:** $L \subseteq L'$

Let A be any integer such that it is divisible by 5 and A = x-y.
So, here we can say that:

$$x-y= 5k \text{ for some } k \in Z \rightarrow x=y+5k$$

So, basically over here we are supposed to prove that:

$$(x-y) = (y+5k,y)$$

Firstly, let us use y diagonal moves from (1,1) if y >= 0, or from (-1,-1) if y <0, in order to get (y,y).

Now, from (y,y) we have to go to (y+5k,y), so in order to do that let us use k steps of (5,0) if k >= 0, or (-5,0) if k<0.

Let us take an example to make things a bit more clear.

Example:

Let us assume that (x,y) = (6,1)

Over here, we have:

- x-y = 5 for k = 3
- From (0,0)→(1,1), let us use one diagonal move and from (1,1)→(6,1), let us use one move of (5,0)

Hence, for (x,y) ∈ L, so over here we can move diagonally to (y,y) and we can also move horizontally (x,y) = (y+5k,y).

Therefore, this just gives a path using allowed moves, so now we can say that L ⊆ L'.